

10) Is the Perceptron able to learn the XOR function? Explain why?
Ans:-

No, a single-layer perceptron is not able to learn the XOR function because the XOR dataset is not linearly separable. A single perceptron can only classify data that can be separated by a single straight line. ~~For~~ Since the XOR inputs cannot be separated by a single decision boundary, the perceptron fails to converge and continuously produces classification errors even after multiple epochs. Therefore, solving the XOR problem requires a Multi-Layer Perceptron (MLP) with one or more hidden layers and non-linear activation functions.